

Anunay Dixit

EDUCATION

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The University of Texas at Austin

Austin, TX

Bachelor of Science in Mechanical Engineering, Minor in Computer Science

May 2026

Relevant Coursework: Mechatronics, Data Structures and Algorithms, Dynamic Systems and Controls, Machine Elements, Dynamics, Engineering Design, Engineering Computations

EXPERIENCE

Argon Mechatronics

June 2025 – August 2025

Robotics Engineering Intern

Austin, TX

- Assembled SCARA robotic arms and validated operational readiness in a fast-paced startup manufacturing environment
- Designed and manufactured jigs to enable repeatable manufacturing iterations across multiple toolhead configurations
- Programmed and executed 50+ G-code routines for multiple toolheads to support automated manufacturing workflows
- Diagnosed mechanical vibration, sensor misalignment, and motor instability, achieving ± 0.045 mm end-effector accuracy
- Documented assembly and machining workflows, logging issues, fixes, G-code changes, and next-step corrective actions

Femto Lab

Oct. 2024 – May 2026

Research Assistant

Austin, TX

- Designed a pinion gear-driven positioning system enabling precise camera motion for layered organoid image acquisition
- Developed a 1 TB+ data transfer system using Raspberry Pi to interface organoid imaging data with cloud storage
- Implemented a Flask-based API to automate data transmission, reducing manual handling and improving efficiency
- Optimized image compression (JPEG, PNG) to reduce storage requirements while preserving analytical image quality

WakeWheel - Awarded Best Project

Sep 2023 – Dec 2023

Tech Engineer

Austin, TX

- Designed and developed an IoT-based driver safety system using ESP32C3 and heartbeat sensors to detect drowsiness
- Implemented C++ control logic and Firebase communication to deliver real-time haptic feedback and React-based alerts

Longhorn Racing

Sep 2022 – Aug 2024

Composites Engineer

Austin, TX

- Evaluated carbon fiber manufacturing methods and part geometries in collaboration with aerodynamics engineers to optimize structural integrity, manufacturability, and overall vehicle performance requirements within a \$5,000 budget
- Manufactured carbon fiber body components using wet layup, vacuum bagging, and oven curing, troubleshooting recurring fabrication defects including vacuum seal failures and ply alignment discrepancies during production cycles
- Iterated layup procedures and tooling to successfully produce required composite components for final vehicle assembly

PROJECTS

Morphing Robotic Hand

Sep 2025 – May 2026

- Partnered with the Minimax Lab to design a morphing robotic hand for integration with an existing KUKA robotic arm
- Designed a fully reconfigurable palm that enables transitions between anthropomorphic and 120° gripper configurations
- Implemented direct-drive servo actuation architecture to achieve controlled motion and reliable torque transmission
- Ensured ROS-compatible control framework for seamless integration with the existing laboratory computing systems

Custom-Built RC Vehicle

Feb 2025 – Apr 2025

- Engineered an RC vehicle integrating drivetrain, steering, and suspension subsystems ensuring performance optimization
- Validated electrical and mechanical performance through iterative prototyping within a \$100 total component budget
- Implemented a 3:1 gear ratio and Ackermann steering geometry, achieving a 0–10 mph acceleration in under 2 seconds

Curie.AI

October 2024

- Built a chatbot-driven platform retrieving 5+ research papers per query via database APIs and suggesting experiments
- Integrated a file upload system and chatbot, enabling users to submit documents and receive responses to their queries

Injection-Molded Handheld Spinner

May 2023

- Fabricated an injection-molded component using CAD and FEA to optimize structural integrity and manufacturability
- Generated CAM toolpaths for CNC machining of aluminum mold tooling, maintaining ± 0.1 mm dimensional tolerance
- Performed injection molding trials to validate proper part fit with bearing caps while minimizing material flow defects

TECHNICAL SKILLS

Manufacturing & CNC: CNC Machining, CAM Toolpath Generation, G-code Programming

CAD Tools: SolidWorks, AutoCAD, Inventor, Fusion 360, GD&T

Languages: Python, Java, Swift, C++, MATLAB, HTML, C#

Other Tools: Flask, Git, Vim, Bash Scripting, Xcode, Microsoft Office